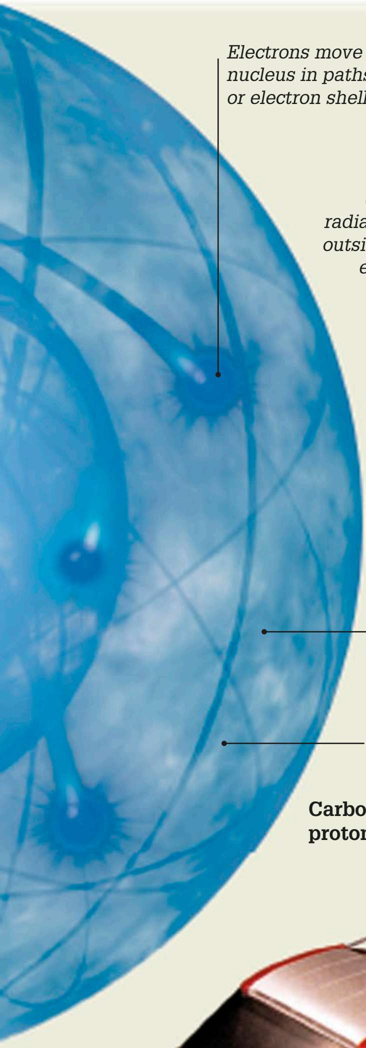
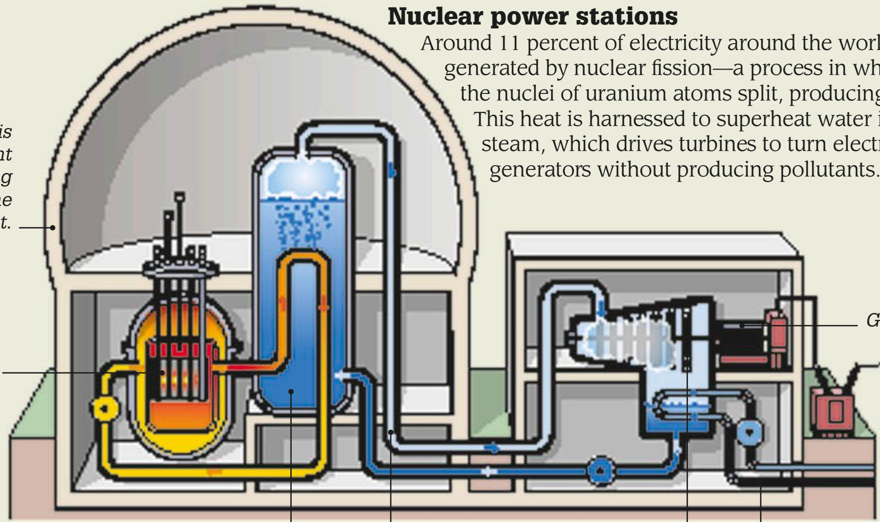
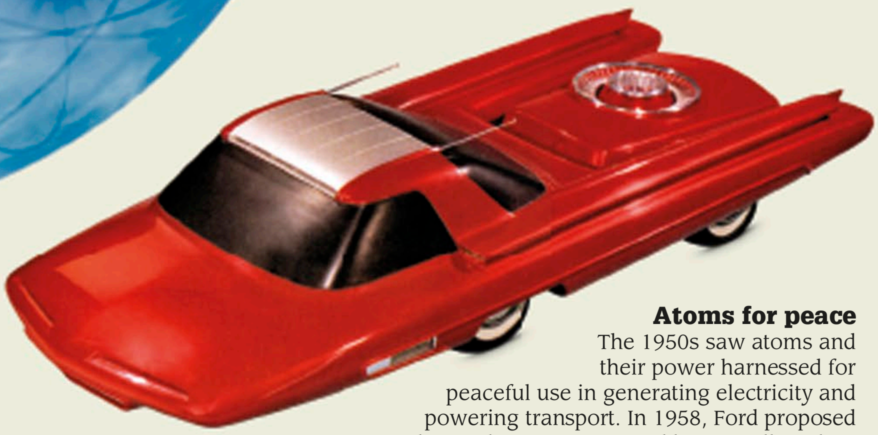


Carbon atom with six protons and electrons

Nuclear power stations

Around 11 percent of electricity around the world is generated by nuclear fission—a process in which the nuclei of uranium atoms split, producing heat. This heat is harnessed to superheat water into steam, which drives turbines to turn electricity generators without producing pollutants.





Model of Ford's Nucleon concept car

Atoms for peace

The 1950s saw atoms and their power harnessed for peaceful use in generating electricity and powering transport. In 1958, Ford proposed the Nucleon car, powered by a small nuclear reactor in its rear. While it was never built, nuclear-powered ships and submarines were made in the following years.

Uses of radioactivity

Sterilization

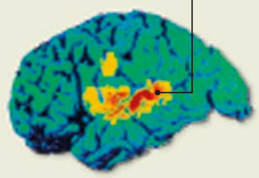
Radiation is used to preserve certain foods as well as sterilize medical instruments, by killing potentially harmful microorganisms and preventing infection.

Dating rocks

Uranium found in many rocks is unstable and decays to lead over time. Measuring the ratio of uranium to lead can help date rocks accurately.

Medicine

A positron emission tomography (PET) scan uses a radioactive chemical, which is injected into the human body and then traced to reveal body activity and diagnose diseases.



PET scan of human brain activity when hearing and repeating words

1932

Using a machine called a particle accelerator, English physicist John Cockcroft and Irish physicist Ernest Walton split the nucleus of an atom—lithium in this case—for the first time ever.

1938

Studies by German chemists Otto Hahn and Fritz Strassmann and Austrian physicist Lise Meitner showed how uranium atoms can be split by nuclear fission to start a nuclear chain reaction.

1954

USS *Nautilus*—the first submarine powered by its own nuclear reactor—was launched. It traveled 347,960 miles (560,000 km) in its first 12 years of operation.

1956

Calder Hall at Sellafield, UK, became the first commercial nuclear power station, producing large quantities of electricity.

USS *Nautilus*